2ND MONITORING REPORT DAK SRONG HYDROPOWER PROJECT

Document Prepared by Kyoto Energy Pte Ltd

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The purpose of the project activity is to generate renewable electricity from a run-of-river hydropower plant. Renewable electricity is exported to the Vietnamese national grid system and, in doing so, the project reduces greenhouse gas emissions. This is achieved by displacing generation by more traditional, fossil fuel dependent methods that lead to anthropogenic emissions of carbon dioxide and other greenhouse gasses (the baseline scenario).

The project is a run-of-river hydropower plant with a small run-of-river reservoir and consists of a weir, a penstock, a powerhouse (containing turbines and generators) and a tailrace. The project has three units each with a turbine generator set with installed capacity of 8 MW – and all three have been operational during thing the monitoring period. The power generated by the project is exported to the Viet Nam national grid through a 110 kV transmission line. Details of the equipment used in the project are provided in section 3.1.

The project was registered as a CDM project (number 3389) on 23-February-2011 and started commercial operations on 30-October-2010. The monitoring period for this report is 01-March-2012 to 22-February-2018.

The project activity has generated a total of 186,159 tCO2e emission reductions during the current proposed VCS monitoring period.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1. Energy industries (renewable - / non-renewable sources)
Project Type: Renewable Energy Projects
The project activity is not a grouped project.

1.3 Project Proponent

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Organization name	Tay Nguyen Hydropower Company Limited

Contact person	Mr. Le Van Dai
Title	Director
Address	3rd Floor, Se San 3A Building, No. 96, Pham Van Dong Street, Hoa Lu Ward, Pleiku City, Gia Lai Province, Vietnam
Telephone	+84 269 222 2466
Email	Vandai.khkt@gmail.com

1.4 Other Entities Involved in the Project

Organization name	Kyoto Energy Pte Ltd
Contact person	Mr. Michel Buron
Title	Director
Address	2 Venture Drive #11-15 Vision Exchange Singapore 608526
Telephone	+33 6 38 94 97 87
Email	Michel.buron@kyotoenergy.net

1.5 Project Start Date

According to VCS Standard, the project start date is the date on which the project began generating GHG emission reductions or removals. Hence for the project, start date is considered as the Commercial Operation Date (COD) which is 30-October-2010.

1.6 Project Crediting Period

The crediting period started on 30-October-2010 and ended on 22-February-2018 (the date when the CDM crediting period expired). This gives a total crediting period of 7 years, 3 months and 22 days. While the original VCS crediting period was 10 years and renewable twice, the final crediting period was determined by the CDM crediting period which was not extended.

1.7 Project Location

(a) Host Party(ies): Socialist Republic of Viet Nam

(b) Region/ State/ Province, etc.: Gia Lai Province,

(c) City/ Town/ Community, etc.: Yang Nam, Ya Ma, and Dak Hninh Communes,Kong Chro district

(d) Physical/ Geographical location: Latitude of 13°41'45"N and longitude of 108° 33'30"E

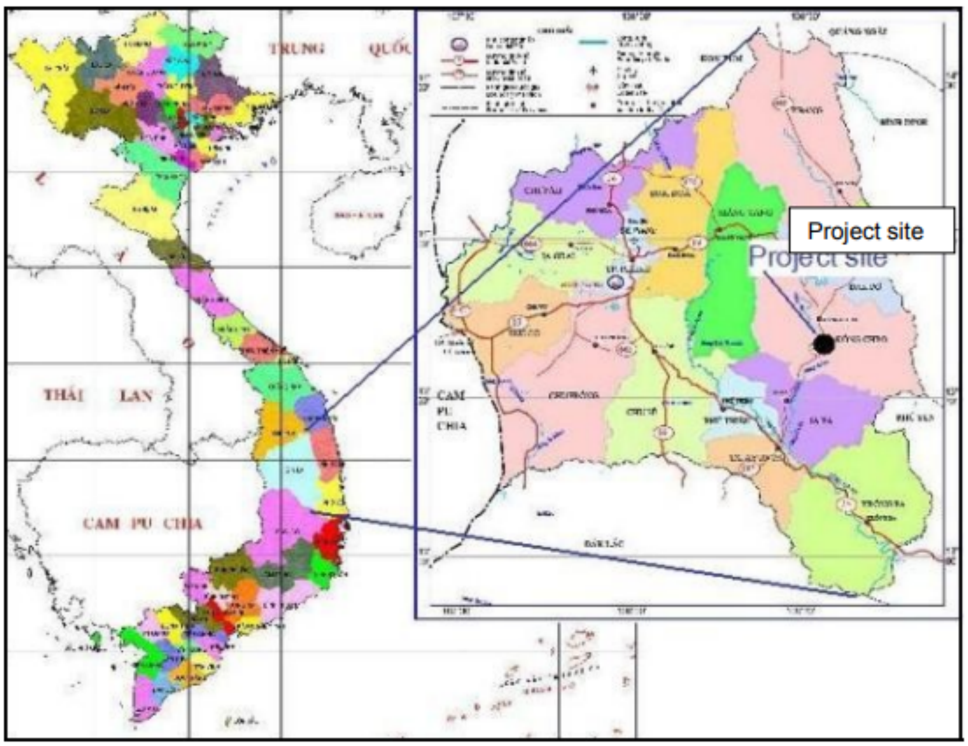

Figure 1: Project Location

1.8 Title and Reference of Methodology

As the project is registered under the Clean Development Mechanism, it employs the methodologies and tools as below:

<u>Methodology:</u>

ACM0002, Consolidated Methodology for Grid Connected Electricity Generation from Renewable Sources, Version 11.

Tools:

- Tool for the demonstration and assessment of additionality (Version 5.2)
- Tool to calculate the emission factor of an electricity system (Version 1.1)

1.9 Participation under other GHG Programs

The project is registered under the Clean Development Mechanism. Details are as below:

Project Title: Dak Srong 2 Hydropower Project

Project Number: 3389

Date of Registration: 23-February-2011

Issuance details:

Date of Issuance: 15-November-2012

Monitoring period: 23-February-2011 to 29-February-2012

Amount of Certified Emission Reductions: 41,560

1.10 Other Forms of Credit

The project does not participate in any other Emission Trading Programs and the GHG emission reductions generated during the current crediting program have not been used for compliance under any programs or mechanisms.

No other forms of GHG related environmental credits have been sought for this project.

1.11 Sustainable Development

An analysis of the economic, social and environmental aspects of the project shows that the project meets the host country's sustainable development criteria for a Clean Development Mechanism project. The project has positive impacts with respect to the environment (offsetting fossil fuel use and lowering greenhouse gas emissions), society (providing jobs, development of roads), technologically (technology transfer) and economy (satisfying growing energy demands to allow the country and region to develop and alleviate poverty). In order to quantify the sustainable development contribution of this project, the project owner has voluntarily agreed to donate a portion of CER revenue from the project towards sustainable development initiatives – for example building and improving infrastructure of the locality and sponsoring social activities of the local communities.

2 SAFEGUARDS

2.1 No Net Harm

As per the Environmental Impact Assessment developed prior to the implementation of this project, the potential environmental and socio-economic impacts of the project were not considered as significant and were mainly temporary (related to the development rather than the operation of the project).

In addition, the project owner follows an environmental monitoring plan and strictly complies with all related Vietnamese laws.

2.2 Local Stakeholder Consultation

The stakeholder consultation meeting was held at the Administrative Building of Kong Ch'ro District People's Committee, Gia Lai Province, Viet Nam at 9:00 AM on 28-July-2008. Personal invitations were sent to community leaders, local People's Committee representatives and placed in the media. Public notices of the planned consultations were placed in National Resources and Environmental Newspaper which is widely published and read in the provinces. Across the consultation, presentations were made by the project owner and consultant who outlined the planned project activity in a non-technical manner (including environmental, social and technological considerations), climate change, the role of the Clean Development Mechanism and annual emission reductions potential. In addition, questionnaires were circulated and filled in by the attendees. In all, there were 20 participants attending the meeting.

Figure 2: Picture of stakeholder meeting in the project site in 28/07/2008

2.3 AFOLU-Specific Safeguards

N/A as this is project does not involve AFOLU.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The project has been fully operational since 30-October-2010. The project has three turbine/generator units of 8 MW each (details are provided in section 1.1).

The electricity generated from Dak Srong 2 hydropower plant is connected to the National Grid through a new 110 kV single circuit transmission line from 110 kV Transformer Station. In practical terms, the transformer of the Dak Srong 2 hydropower plant is connected to the 110 kV transformer station of the Dak Srong 2 A hydropower plant which is in turn connected to the 110 kV transformer station of Dak Srong hydropower plant which is ultimately connected to the National Grid. The length of transmission line belonging to this project is 10.2 km..

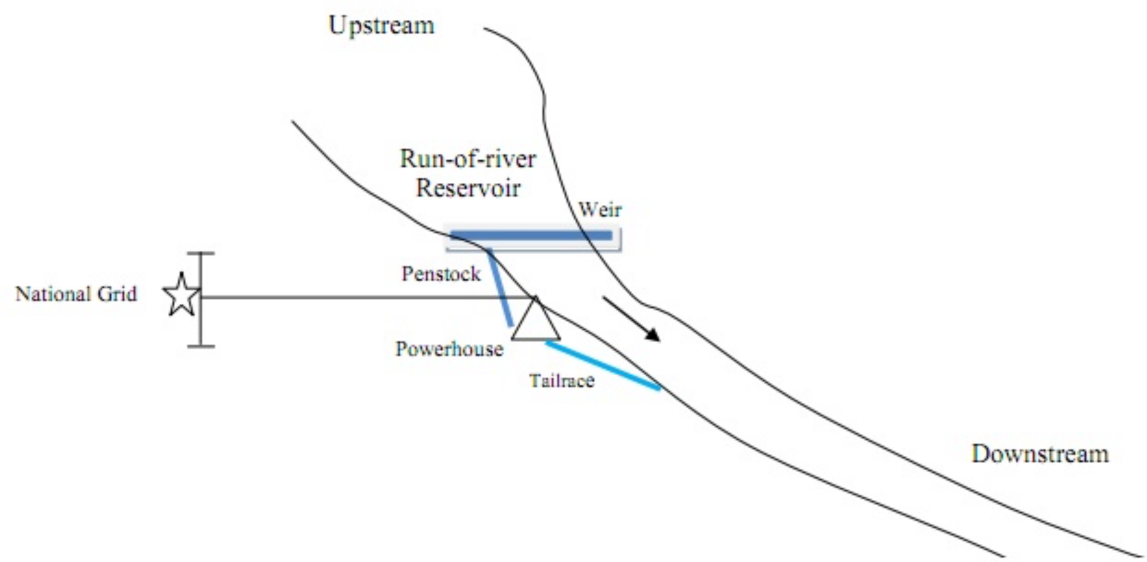

Figure 3: Project overview

The project activity has a total installed capacity of 24MW, made up of 3 units each with an installed capacity of 8MW.

The key equipment utilized in the project is described in the table below:

Table 1: *Key technologies utilized*

	Item	Specification
Turbines	Quantity	3
	Capacity	8,290.2 kW
	Type	HLA551C-LJ-176
	Design head	37.5
	Rated speed	300 rpm
	Rated efficiency	93.5

	Turbine discharge at design head	24.11 m^3/s
Generators	Quantity	3
	Capacity	8,000 kW
	Type	SF 8,000-20/3250
	Cosα	0.8
	Rated speed	300 rpm
	Runaway speed	527.6 rpm
	Rated voltage	6.3 kV
	Rated efficiency	96.50%

No specific events have taken place during this monitoring period.

3.2 Deviations

3.2.1 Methodology Deviations

None

3.2.2 Project Description Deviations

None

3.3 Grouped Projects

N/A

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,CM,y}$

Data unit	tCO_2/MWh
Description	CO_2 emission factor in year y Combined margin CO_2 emission factor for grid connected power generation in year y calculated using “Tool to calculate the emission factor for an electricity system” (version 01).
Source of data	As per GEF report from Vietnam DNA “Study, Definition of Viet Nam Grid Emission Factor, 2009”
Value applied	0.5764
Justification of choice of data or description of measurement methods and procedures applied	As documented in registered Project Design Document
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF_{Res}
Data unit	kg CO2/MWh
Description	Default emission factor for emission from reservoirs
Source of data	Decision by EB23
Value applied	90
Justification of choice of data or description of measurement methods and procedures applied	As per methodology
Purpose of Data	Calculation of baseline emissions
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid during monitoring period
Source of data	Project activity site
Description of measurement methods	Bidirectional electricity meters

and procedures to be applied	
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value monitored	384,854
Monitoring equipment	Main meters: MM1, MM2, MM3; Backup meters: BM1,BM2, BM3 in case of the MM failure See Appendix 1 for details.
QA/QC procedures to be applied	Cross check measurement results with records for sold and purchased electricity. Meters calibrated at least every two years[1].
Purpose of the data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	TEG_y
Data unit	MWh
Description	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads
Source of data	Electricity meters
Description of measurement methods and procedures to be applied	Measured individually at energy meters TEG1, TEG2 and TEG3. Total is calculated as the sum of the three values
Frequency of monitoring/recording	Continuous measuring Monthly reading Monthly recording
Value monitored	396,342
Monitoring equipment	TEG1, TEG2, and TEG3. See Appendix 1
QA/QC procedures to be applied	Meters calibrated every two years or less.
Purpose of the data	Calculation of project emissions

[1] Following the first CDM verification, calibration frequency of once in two years as per decision no. 25/2007/QD-BKHCN has been adopted.

Calculation method	TEG = TEG1 + TEG2 + TEG3
-	-

Data / Parameter	CAP_{BJ}
Data unit	W
Description	Installed capacity of the hydro power plant after the implementation of the project activity
Source of data	Project Site
Description of measurement methods and procedures to be applied	Determine the installed capacity based on recognized standards (Name plates of generating equipment)
Frequency of monitoring/recording	Yearly
Value monitored	24,000,000
Monitoring equipment	N/A
QA/QC procedures to be applied	-
Purpose of the data	Calculation of project emissions
Calculation method	Not applicable
Comments	-

Data / Parameter	A_{PJ}
Data unit	M^2
Description	Surface area of the reservoir measured after the implementation of the project activity, when the reservoir is full (m2)
Source of data	Project Site
Description of measurement methods and procedures to be applied	Water level measured on site a water level gauge, and surface area is calculated by area to water level chart (made specifically by a third party consultant for the reservoir associated with the project activity and based on topographical maps).
Frequency of monitoring/recording	Annual measurement
Value monitored	4,550,000

Monitoring equipment	N/A
QA/QC procedures to be applied	-
Purpose of the data	Calculation of project emissions
Calculation method	Not applicable
Comments	-

4.3 Monitoring Plan

4.3.1 Description of the monitoring system

The system is composed of six energy meters (3 main and 3 backups) belonging to the grid operator EVN and used as the source of data for the calculation of emission reductions.

There are also three meters (TEG), one located after each generator, which measure the total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads.

The location of each instrument is indicated schematically in the monitoring diagram shown in Figure 4:

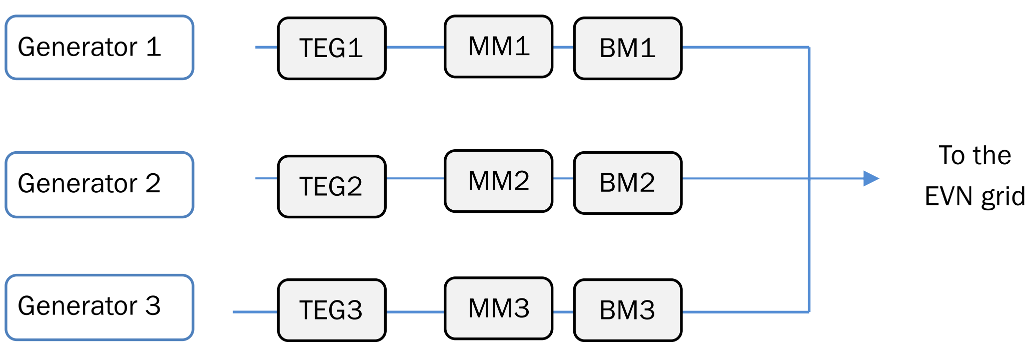

Figure 4: Schematic representation of the meters location on the electric circuit

- MM1, MM2 and MM3 are the main meters. They measure the electricity generation at the hydropower site while taking into account both export and import to/from the national grid. The MMs are the source of data for the EVN receipts.
- BM1, BM2 and BM3 are the backup meters. They are used to be compared the records from the main meters, and as record source in case at least one the main meter fails.

- TEG1, TEG2 and TEG3 are the meters located at after the Generator 1, Generator 2 and Generator 3 respectively. They measure the total electricity generation of each generator. They are used to calculate the project emissions.

4.3.2 Organizational structure, responsibilities, and competencies.

The monitoring team is composed of the following staff:

Report to:	Position
Tay Nguyen Hydropower Company Limited	Site manager
	Site supervisor
	Operator Head of Shift
	Operators (3 per shift)
Kyoto Energy Pte Ltd	Consultant/ Project manager

Table 2: CDM Monitoring Team Details

#	Tasks description (and frequency)	Operator	Supervisor	Site manager / Project director	Consultant Project manager
Monitoring activity					
1	Recording of manual data	✓[2]			
Quality Assurance & Quality Control					
2	Verification of data monitored (consistency and completeness)		✓		
3	Ensuring adequate training of staff		✓		
4	Ensuring adequate maintenance		✓		
	Ensuring calibration of monitoring instruments		✓		
5	Data archiving: ensuring adequate storage of data monitored (integrity and backup): 2 years after the end of the crediting period			✓	
6	Identification of non-conformance and corrective/preventive actions and monitoring plan improvement		✓		
7	Emergency procedures		✓		
8	External audit				✓

[2] This is a joint reading with representatives from the power company, as per Power Purchasing Agreement

#	Tasks description (and frequency)	Operator	Supervisor	Site manager / Project director	Consultant Project manager
Calculation of GHG emission reductions and reporting					
9	Processing of data and calculation of emission reductions			✓	
10	Monitoring report: management review of monitoring report (internal audit)			✓	

Table 3: Allocation of monitoring responsibilities

4.3.3 Description of methods for generating, recording, storing, aggregating, collating and reporting data on monitored parameters

The log files are kept for a minimum of 2 years after the end of the last crediting period by using paper documents and electronic files.

The log files are stored on various media (CD-ROM and hard disks) at several locations (plant, headquarters and consultant server).

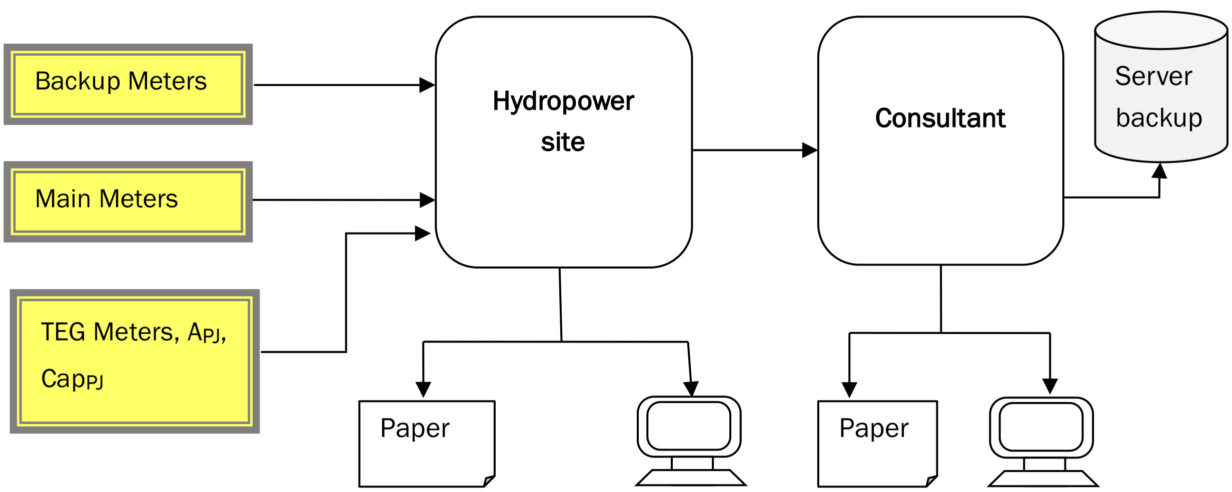

Figure 5: Monitoring data flow chart

The daily internal record procedure is as follows: Everyday at midnight the operator on duty takes the current reading of the 3 main meters and inputs these values into the excel logfile on the site computer.

The preparation of the EVN receipt is as follows: At the end of every month, representatives of the project and of Central Power Company come to the site. The meters keep in their memory the values recorded at the end of the last day of previous month, so that they can take a joint reading and prepare the receipt together. Readings are taken for both main and backup meters. The sale receipt prepared at the site serves as basis for the payment.

The TEG internal record procedure is as follows: Every last day of each month at midnight, the operator on duty takes the current reading of the 3 main meters and inputs these values into the excel logfile on the site computer.

The external audit was performed by the consultant for the crediting period. The internal audit of the monitoring report was performed by the responsible entity of the project participant.

4.3.4 Internal auditing and non-conformities

Internal auditing:

- **Data treatment and verification:**
  A verification of the data recorded was performed by:
  - The spot check of field instruments by staff; and
  - The comparison between the main meters and backup meters (the deviation between MM1 and BM1, MM2 and BM2, MM3 and BM3).

- **Maintenance and calibration of monitoring instrument:**
  Calibration takes place to ensure that the monitoring equipment is correctly installed and functioning properly.

4.3.5 Non-conformities:

Non conformities are internally defined as any incidents affecting the project's monitoring. (e.g.: calibration delay, data loss, instrument malfunction, change in project implementation, etc.) They feature various severities and some of them may lead to data reconstruction (emergency procedure), request for deviation or revision of the monitoring plan, whereas some others may not require any particular corrective measures (e.g.: temporary grid outage).

In the event that the metering system suffers any failure, damage and unexpected problems, or if any errors in the main metering systems are detected during calibration, the electricity exported will be identified as follows.

- Using the results of the backup system

- Should the backup system also suffer a breakdown, the electricity exported is proposed by reconstructing data by means of trend analysis (taking a conservative approach)

The above is also in line with the procedures stipulated in the power purchase agreement (PPA).

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$ Since: $EG_{PJ,y} = EG_{facility,y}$ Hence: $BE_y = EG_{facitity,y} * EF_{grid,CM,y}$			
Where:	Description	Units	ID/value
BE_y	Baseline Emissions	tCO_{2e}	221,830
$EG_{facitity,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid during the current monitoring period	MWh	384,854
$EF_{grid,CM,y}$	CO_2 emission factor in year y	tCO_{2e}/MWh	0.5764

5.2 Project Emissions

Power density: $PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$			
Where:	Description	Units	ID/value
PD	Power density of the project activity	W/m^2	5.27

Cap_{PJ}	Installed capacity of the hydropower plant after the implementation of the project activity	W	24,000,000
Cap_{BL}	Installed capacity of the hydropower plant before the implementation of the project activity (W). For new hydropower plants, this value is zero	W	0
A_{PJ}	Area of the reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoirs are full	m^2	4,550,000
A_{BL}	Area of the reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoirs are full (m^2). For the new reservoir, this value is zero	m^2	0

Because the power density of the project is greater than 4 W/m2 and less than or equal to 10 W/m^2, the project emissions are calculated as follows:

Project emissions: $PE_{HP,y} = \frac{EF_{REs} * TEG_y}{1000}$			
Where:	Description	Units	ID/value
$PE_{HP,y}$	Emission from reservoir expressed as	tCO_{2e}/year	35,671
EF_{Res}	The default emission factor for emissions from reservoirs	kg CO_{2e}/MWh	90
TEG_y	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads during the current monitoring period	MWh	396,342

5.3 Leakage

As per the registered methodology, no leakage emissions are to be considered.

$$LE_y = 0$$

5.4 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
2012	26,434	4,245	0	22,189
2013	46,874	7,517	0	39,357
2014	27,857	4,486	0	23,371
2015	23,558	3,796	0	19,762
2016	35,379	5,650	0	29,729
2017	55,271	8,867	0	46,403
2018	6457	1109	0	5,348
Total	221,830	35,671	0	186,159

The ex-ante emission reductions were estimated (at the time of registration) as 44,466 tCO_2e per annum, which would imply emission reductions of 266,065 tCO_2e over the current monitoring period of 2184 days.

On an annual basis, the emission reductions have been lower than expected in all years except for 2017. Across the monitoring period, the end result is an underperformance of ~30%. This is not surprising, given that the ex ante projections are based on a probabilistic forecast of generation based on simulated flow conditions which in turn is largely dependent on rainfall. While in the long term, the output of the project is expected to approach the average, in the short and medium terms deviations are expected – particularly in the case of run of river projects with negligible storage such as this project. The higher performance in 2017 (~4%) is minimal and corresponds to an unexceptional plant load factor of ~46%.

APPENDIX 1: INSTRUMENT MANIFEST

Meter Id	Tag	Model, Serial number and manufacturer	Accuracy Class	Date of Initial Calibration (prior to calibration period)[3]	Date of Subsequent Calibration	Validity Date of Calibration	Event
Main meter	MM1	Manufacturer: Elster Model: A1700 Serial: 10020491	0.2S	25/07/2011	03/07/2012 02/08/2013 23/07/2014	03/07/2014 02/08/2015 23/07/2016	Meter changed on 22/08/2015
Main Meter	MM1	Manufacturer: Elster Model: A1700 Serial: 10125999	0.2S	17/07/2015	25/07/2016 24/07/2017 28/07/2018	25/07/2018 24/07/2019 28/07/2020	
Main meter	MM2	Manufacturer: Elster Model: A1700 Serial: 10020492	0.2S	05/07/2011	03/07/2012	03/07/2014	Meter replaced on 10/07/2013
Main Meter	MM2	Manufacturer: Elster Model: A1700 Serial: 10020490	0.2S	10/07/2013	23/07/2014	23/07/2016	Meter replaced on 31/08/2015
Main Meter	MM2	Manufacturer: Elster Model: A1700 Serial: 10020493	0.2S	31/08/2015	-	-	Meter replaced on 23/07/2016
Main Meter	MM2	Manufacturer: Elster Model: A1700 Serial: 15014070	0.2S	23/07/2016	-	-	Meter replaced on 12/08/2017
Main Meter	MM2	Manufacturer: Elster Model: A1700 Serial: 17029694	0.2S	19/07/2017	28/07/2018	28/07/2020	

[3] Calibration frequency is 2 years as per decision no. 25/2007/QD-BKHCN, dated 05/10/2007

Main Meter	MM3	Manufacturer: Elster Model: A1700 Serial: 10020493	0.2S	25/07/2011	-	-	Meter replaced on 17/08/2012
Main Meter	MM3	Manufacturer: Elster Model: A1700 Serial: 12012902	0.2S	20/08/2012	02/08/2013 23/07/2014 03/08/2015 25/07/2016 24/07/2017 28/07/2018	02/08/2015 23/07/2016 03/08/2017 25/07/2018 24/07/2019 28/07/2020	
Backup Meter	BM1	Manufacturer: Elster Model: A1700 Serial: 10018220	0.5S	25/07/2011	03/07/2012 02/08/2013 23/07/2014	03/07/2014 02/08/2015 23/07/2016	Meter replaced on 22/08/2015
Backup Meter	BM1	Manufacturer: Elster Model: A1700 Serial: 14163278	0.5S	17/08/2015	25/07/2016 24/07/2017 28/07/2018	25/07/2018 24/07/2019 28/07/2020	
Backup Meter	BM2	Manufacturer: Elster Model: A1700 Serial: 10018221	0.5S	25/07/2011	03/07/2012 02/08/2013 23/07/2014	03/07/2014 02/08/2015 23/07/2016	Meter replaced on 22/08/2015
Backup Meter	BM2	Manufacturer: Elster Model: A1700 Serial: 14163279	0.5S	17/08/2015	25/07/2016 24/07/2017 28/07/2018	25/07/2018 24/07/2019 28/07/2020	
Backup Meter	BM3	Manufacturer: Elster Model: A1700 Serial: 10018222	0.5S	25/07/2011	16/07/2012 02/08/2013 23/07/2014	16/07/2014 02/08/2015 23/07/2016	Meter Replaced on 22/08/2015
Backup Meter	BM3	Manufacturer: Elster Model: A1700 Serial: 14163280	0.5S	17/08/2015	25/07/2016 24/07/2017 28/07/2018	25/07/2018 24/07/2019 28/07/2020	
TEG	TEG1	Manufacturer: Wasion Model: DSSD331	0.5S	10/10/2010	10/10/2012 10/10/2014	10/10/2014 10/10/2016	

		Serial: 09090155620006			15/10/2016[4] 12/10/2018	15/10/2018 12/10/2020	
TEG	TEG2	Manufacturer: Wasion Model: DSSD331 Serial: 09090155620007	0.5S	10/10/2010	10/10/2012 10/10/2014 15/10/2016[5] 12/10/2018	10/10/2014 10/10/2016 15/10/2018 12/10/2020	
TEG	TEG3	Manufacturer: Wasion Model: DSSD331 Serial: 09090155620008	0.5S	10/10/2010	10/10/2012 10/10/2014 15/10/2016[6] 12/10/2018	10/10/2014 10/10/2016 15/10/2018 12/10/2020	

[4] Due to delay in calibration, a penalty (increase) of 0.5% has been applied to the total generation for the month of October 2016

[5] Due to delay in calibration, a penalty (increase) of 0.5% has been applied to the total generation for the month of October 2016

[6] Due to delay in calibration, a penalty (increase) of 0.5% has been applied to the total generation for the month of October 2016